

PATIENT

We have to develop a web app called Priori. Our website is a **pre-arrival hospital triage and intake system** that helps patients get prioritized *before* they reach the hospital. Instead of waiting in long, unorganized queues, users enter their symptoms (manually or via voice), and the system analyzes them using a rule-based triage engine to assign a priority level. Based on this, patients are placed into a **real-time, dynamically updated queue**, and given an estimated waiting time rather than a fixed appointment. The platform also suggests nearby hospitals based on distance and current load, while allowing users to choose. On the hospital side, staff get a **live dashboard of incoming patients**, sorted by urgency, along with explainable insights to support faster and better decision-making. Overall, the system reduces delays, improves patient prioritization, and helps hospitals prepare in advance for incoming cases.

Common features in all pages:

- All the page will have dark/light theme options
- Top profile logo thing with logout option

Page 1: Login/Sign up

At the top it has "Welcome to PRIORI"

This has 2 user end one by patient and another hospital admin

Login page with a sign-up option, depending on user's choice it will take you to the patient dashboard or the admin dashboard. When a new patient sign's up they are required to give name, phone number, mail id, set a password and it is optional for them to set up Medical history(optional during sign up, quick/basic info) while signing up.

Admin name admin phone no n email id with hospital id are absolutely necessary while signing up

After login all patient pages should have the following panel on left side:

- **Dashboard**
- **Symptom Filling**
- **Hospital Suggestions**
- **Family Profiles**
- **Medical History**
- **Alerts**
- **Feedback system**
- **Profile**

When u tap into that it takes u to that particular page

Page 2: Patient dashboard

Top: "Welcome, [User Name]"

4 Placards (Cards)- right side of the panel:

- Symptom Check
 - Display symptoms and hyperlink to the Symptom Check page
- Hospital Suggestions
 - Display suggested hospitals and thyperlinks to Hospital Suggestions page
- Live Status
 - Live Status (EDA / ETA)
 - This is system-calculated, not a message.
 - What it shows: Estimated wait time (range), "12–18 mins", Continuously updating
 - Based on: queue, doctor availability, priority shifts
 - This is predictive + numeric + dynamic
- Alerts
 - This is human/system-triggered communication
 - What it shows: Status messages: "Preparing for your arrival", "Doctor has been notified"
 - Event updates: "Ambulance dispatched", "Delay due to emergency case", "You are next"
 - This is informational + event-based + discrete

FLOATING ACTION

Emergency SOS Button (Bottom-right corner, Visible on all pages)

Feature: Emergency SOS Button (Patient Side)

Build a floating Emergency SOS button fixed at the bottom-right corner of all patient pages.

UI Behaviour:

- Red button with SOS written on it
 - Always visible (z-index high)
 - On click → open modal
-

Modal Flow:

Ask the user:

1. **Do you need an ambulance?**
 - Yes / No
2. **What happened? (optional)**
 - Text input (short description)
3. **Extra requirements (optional):**
 - Oxygen
 - Wheelchair
 - Other (text)
4. **Confirm SOS**

On Confirm:

1. Get User Location (FREE)

Use browser API:

```
navigator.geolocation.getCurrentPosition()
```

Store:

- latitude
- longitude

2. Select Hospital (NO PAID API)

- Use pre-stored hospital list (hardcoded JSON or DB)
- Each hospital has:
 - name
 - lat/Ing
 - contact number
- Calculate nearest hospital using simple distance formula (Haversine)

3. Send Alert to Backend

POST `/api/sos`

Payload:

```
{
  "userId": "...",
  "location": { "lat": ..., "Ing": ... },
  "description": "...",
  "ambulance": true,
  "requirements": ["oxygen"],
  "hospitalId": "..."
}
```

}

4. Notify Hospital Admin (FREE OPTIONS)

Option A (Recommended for hackathon):

- Use **Firestore Cloud Messaging (FCM)** (free)
- Send push notification to hospital dashboard

Option B (Simpler demo):

- Show real-time alert via **Socket.io**
- Hospital dashboard instantly receives SOS

5. Optional SMS (if needed)

- Use free-tier like **Twilio trial**
OR
- Skip SMS and simulate in demo

Hospital Side Behavior:

- SOS appears instantly on dashboard:
 - Patient name
 - Location
 - Emergency details
- Highlight as HIGH PRIORITY

Notes:

- Do NOT depend on paid APIs (Google Maps etc.)
- Keep hospital list static for demo
- Prioritize speed + reliability over complexity
- If location fails → fallback message: "Location unavailable"

Expected Output:

- One-click emergency flow
- Real-time alert on hospital side
- No page reload
- Works within free-tier tools only

Page 3: Symptom Check

Overview

The Symptom Check page allows users to quickly select their primary symptoms and answer a few targeted follow-up questions. The goal is to collect structured, reliable input for the system to process and generate an accurate estimated wait time (ETA).

The interface is designed to be **fast, intuitive, and non-overwhelming**, using selectable cards (placards) instead of long forms.

Step 1: Symptom Selection (Placards)

Users are shown multiple selectable symptom cards:

- Chest Pain
- Breathing Difficulty
- Fever
- Injury / Accident
- Headache
- Vomiting
- Other

Behavior:

- User can select **one primary symptom**
 - Selected card gets highlighted
 - On selection → **targeted questions appear dynamically below**
-

Step 2: Targeted Questions (Dynamic & Optional)

Based on the selected symptom, 2–4 relevant questions are shown.

Example: Chest Pain

- Type of pain:
 - Sharp / Stabbing
 - Pressure / Tightness
 - Duration:
 - < 5 minutes
 - 5–30 minutes
 - 30 minutes
 - Additional symptoms:
 - Sweating? (Yes / No)
 - Pain spreading to arm or jaw? (Yes / No)
 - Difficulty breathing? (Yes / No)
-

Example: Fever

- Temperature level:
 - Mild
 - High
 - Duration:
 - 1 day
 - 2–3 days
 - More than 3 days
 - Additional symptoms:
 - Chills? (Yes / No)
 - Weakness? (Yes / No)
-

Behavior:

- All targeted questions are **optional**
 - Keeps interaction fast and reduces user friction
-

Step 3: Additional Symptoms (Optional Input)

- Text field:
 - “Is there anything else you would like to tell us?”
- Allows users to mention extra symptoms not covered above

Step 4: Condition Check (Severity Anchoring – Optional)

Instead of asking users to rate pain numerically, simple functional questions are used:

- Can you walk normally?
- Are you able to speak properly?
- Are you feeling dizzy?

Behavior:

- All questions are optional
 - Helps improve accuracy without overwhelming the user
-

Step 5: Actions

Two buttons are provided:

1. Submit

- Sends the collected data to the backend triage system
- System processes symptoms and generates:
 - Estimated wait time (ETA)
 - Internal priority level (not shown to user)
- Creates an **Active Case**

👉 After submission, the user is redirected to the **Hospital Suggestions page**

2. Edit Symptoms

- Allows user to:
 - Change selected symptom
 - Modify answers
 - Does not trigger submission
-

Flow Summary

1. User selects a symptom

2. Targeted questions appear
 3. User optionally fills extra details
 4. User optionally answers condition questions
 5. User clicks **Submit**
 6. System processes input
 7. User is redirected to **Hospital Suggestions page**
-

Key Design Principles

- No long forms
 - Minimal typing
 - Maximum structured input
 - Fast interaction (few taps)
 - No medical jargon
 - No explicit mention of “triage”
-

Important Note

- Symptom submission happens **before hospital selection**
- The system first processes the case, then suggests hospitals
- This ensures:
 - Better hospital recommendations
 - More accurate ETA

Page 4: Hospital suggestions

Hospital Selection – Data Routing to Admin

Overview

Once the user selects and confirms a hospital, all patient details are sent **only to the selected hospital's admin system**.

This ensures that:

- Hospitals receive only relevant patient data
 - Patient information is not broadcast to multiple hospitals
 - The selected hospital can prepare in advance
-

Data Sent to Hospital

After the user clicks “**Confirm Hospital**”, the system creates/updates an **Active Case** and sends the following data:

Patient Details

- Name
 - Age (if available)
 - Contact number
-

Symptom Data

- Selected primary symptom
 - Answers to targeted questions
 - Additional symptoms (text input)
 - Condition/functional responses (walking, speaking, dizziness)
-

Medical Context

- Medical history (if available):
 - BP
 - Diabetes
 - Allergies
-

System-Generated Data

- Estimated wait time (ETA)
 - Internal priority level (not shown to patient)
 - Timestamp of submission
-

Location Data

- User's live location (latitude, longitude)
-

Routing Logic

- Data is sent **only to the selected hospital**
- Each hospital has a unique identifier (`hospitalId`)
- Backend ensures:

Send data → only to matching `hospitalId`

- No other hospital receives this data
-

Hospital Admin Side Behavior

Once data is received:

- Patient appears in the **live dashboard**
- Entry includes:

- Patient details
 - Symptoms
 - ETA
 - Location
 - Patient is:
 - Added to the queue
 - Sorted based on severity (internal logic)
-

Real-Time Updates

- Any updates from the patient (e.g., “Update Condition”)
 - are pushed to the **same selected hospital only**
 - SOS alerts (if triggered)
 - are also sent to the **selected or nearest hospital**
-

Key Principle

👉 “One patient → One hospital → One active data flow”

This ensures:

- Privacy
- Clarity
- Efficient hospital preparation

Page 5: Family Profiles (Dependents / Caregiver Use)

Overview

The Family Profiles feature allows users to add and manage details of other individuals (family members, friends, or dependents) and perform symptom check and hospital selection on their behalf.

This is useful in situations where:

- A parent is filling details for a child
 - A user is helping a friend
 - A caregiver is assisting an elderly or dependent patient
-

Adding a Profile

When adding a new profile, the system collects the following details:

Basic Information

- Name of the person
 - Age (or age group)
 - Gender (optional)
-

Relationship

- Relationship with the user:
 - Parent
 - Child
 - Sibling

- Friend
 - Patient (for caregiver use)
 - Other
-

Contact Details

- Phone number of the person (optional)



Note:

The primary contact for all communication (alerts, updates, etc.) remains the **logged-in user**, unless specified otherwise.

Medical Information (Optional)

- Known conditions:
 - Blood pressure (BP)
 - Diabetes
 - Allergies
 - Any additional medical notes (optional)
-

Using a Profile

- The user selects a profile from the Family Profiles section
 - Once selected, the system redirects to the **Symptom Check page**
 - All entered symptoms and data are recorded for the **selected individual**, not the logged-in user
-

Key Behavior

- Each profile acts as a **separate patient identity** within the same account
 - All actions (symptom entry, hospital selection, SOS) apply to the selected profile
 - Notifications and updates are sent to the **account holder by default**
-

Purpose

This feature ensures that users can easily assist others who:

- Cannot use the system themselves
- Require immediate help
- Are under someone's care

Page 6: Medical History Page

Overview

The Medical History page allows users to view and update their health information at any time.

Basic medical details may be collected during signup, but this page provides a **complete and editable view** that helps improve system accuracy during symptom evaluation.

All fields on this page are **optional** and can be updated anytime.

UI Design Approach

- Use **placards / selectable options** for quick input
 - Provide **text fields for additional details**
 - Always include a **“No” option** to avoid forced input
 - Keep the interface simple and non-overwhelming
-

Sections

1. Existing Medical Conditions

Questions:

- Do you have diabetes?
 - Yes / No
- Do you have blood pressure issues?

- No
 - Low BP
 - High BP
 - Do you have cholesterol issues?
 - No
 - Low
 - High
 - Any known heart conditions?
 - Yes / No
-

Additional Input:

- Text field:
 - “Add any other medical conditions”
-

2. Allergies

- Do you have any known allergies?
 - Yes / No

If Yes:

- Text field:
 - “Specify allergies (food, medicine, etc.)”
-

3. Family Medical History

Questions:

- History of heart disease in family?
 - No
 - Yes – Paternal side
 - Yes – Maternal side
 - Yes – Both
 - History of diabetes in family?
 - No
 - Yes – Paternal
 - Yes – Maternal
 - Yes – Both
-

Additional Input:

- Text field:
 - “Any other family medical history”
-

4. General Health Information (Optional but Useful)

- Are you currently on medication?
 - Yes / No

If Yes:

- Text field:
 - “List medications”
-

- Have you had any major surgeries?
 - Yes / No

If Yes:

- Text field:
 - “Describe surgery”
-

5. Notes / Additional Information

- Text field:
 - “Anything else you would like us to know”
-

Actions

- **Save / Update**
 - Stores the medical history
-

Usage in System

- Medical history is automatically used during:
 - Symptom evaluation

- ETA calculation
 - Priority determination (internal)
-

Key Principles

- All inputs are optional
- No complex medical jargon
- Fast to fill and easy to update
- Supports both quick selection and detailed input

Page 7: Alerts

Overview

The Alerts page displays real-time updates sent from the hospital to the patient regarding their case status.

These alerts are generated by the **hospital admin/system** and keep the patient informed about what is happening before and during their visit.

Purpose

- Provide real-time communication
 - Reduce uncertainty and anxiety
 - Keep the patient informed about delays, progress, and readiness
-

Types of Alerts

Alerts may include:

Status Updates

- “Preparing for your arrival”
 - “Doctor has been notified”
 - “You are next”
 - “Doctor will be available in 10 minutes”
-

Emergency / Delay Updates

- “Delay due to emergency case”
 - “High-priority patient being handled”
 - “Slight delay in consultation”
-

SOS / Ambulance Updates

- “Ambulance dispatched”
 - “Ambulance arriving shortly”
 - “Medical team is ready”
-

UI Layout



Top Section

- Title: “**Alerts**”
 - Subtext:
 - “Real-time updates from your selected hospital”
-



Alerts Feed

- Display alerts in a **chronological list (latest first)**

Each alert item shows:

- Message text
 - Timestamp (e.g., “2 mins ago”)
-

Example:

- “Ambulance arriving shortly” — 1 min ago
 - “Preparing for your arrival” — 3 mins ago
 - “Doctor has been notified” — 5 mins ago
-



Real-Time Behavior

- Alerts update automatically (via WebSockets / real-time system)
 - No page refresh required
-

Dashboard Integration

- The **latest alert** is also shown in the **Alerts placard on the dashboard**
 - Any new alert:
 - Updates instantly on dashboard
 - Appears at the top of Alerts page
-

Important Behavior

- Alerts are linked to the **selected hospital only**
 - Each patient receives alerts specific to their own case
 - No alerts from other hospitals are shown
-

Key Principle

 “Alerts tell the user what is happening right now”

- Short
 - Clear
 - Actionable
-

What to Avoid

- Long paragraphs
 - Medical jargon
 - Too many unnecessary alerts
-

Summary

The Alerts page acts as a **live communication channel** between the hospital and the patient, ensuring transparency and improving the overall experience.

Page 8: Feedback

Overview

The Feedback page collects user input about their experience with the system. This helps improve usability, accuracy, and overall performance over time.

Purpose

- Understand user experience
 - Identify issues (complexity, delays, confusion)
 - Improve system flow and triage accuracy
-

UI Design

- Minimal and quick to complete
 - Mostly **tap-based options (placards/buttons)**
 - Optional text input for detailed feedback
-

Feedback Questions

1. Overall Experience

- How was your experience?
 - Excellent
 - Good
 - Average
 - Poor
-

2. Ease of Use

- Was the system easy to use?
 - Yes
 - Somewhat
 - No
-

3. Time Taken

- Did the process feel too long?
 - No, it was quick
 - It was okay
 - Yes, it was too lengthy
-

4. Clarity

- Were the questions clear and easy to understand?
 - Yes
 - Somewhat
 - No
-

5. Helpfulness

- Did the system help you make a decision faster?
 - Yes
 - Not sure
 - No
-

Additional Feedback (Optional)

- Text field:
 - “Any suggestions or issues you faced?”

Action

- **Submit Feedback**
 - Stores responses in database

Backend Usage

- Feedback is saved and used for:
 - Improving question flow
 - Adjusting system logic
 - Enhancing user experience

Key Principles

- Keep it short
- Keep it optional
- No friction
- No mandatory long answers

Important Note

- Feedback should ideally be shown:
 - After hospital visit
 - Or after session completion

Page 9: My Profile

Overview

The Profile page allows users to view and update their personal details and provides a unique QR code for quick identification at the hospital.

This page acts as the user's **identity and check-in interface**.

User Details

Users can view and edit the following:

- Name
- Phone Number
- Email ID

Behavior:

- Fields are editable
 - Changes can be saved anytime
-

QR Code (Hospital Check-In)

A unique QR code is generated for each user.

Purpose:

- Used at hospital entry for quick identification
 - Scanned by hospital admin/staff
 - Instantly retrieves:
 - Patient details
 - Medical history
 - Current case (symptoms, ETA, etc.)
-

UI Behavior:

- QR code is clearly visible
 - Button: “**Show QR**” / “**Enlarge QR**”
 - Optional: Download QR
-

How It Works

1. User arrives at hospital
2. Admin scans QR code
3. System fetches user data instantly
4. No need for manual registration

👉 This reduces waiting time and speeds up intake

Additional Sections

Medical History

- Shortcut to view/edit medical history
-

Past Visits

- List of previous cases:
 - Date
 - Hospital
 - Symptoms (basic)

Key Principles

- Fast access
- Minimal fields
- Easy updates
- Seamless hospital integration

Important Note

- QR should map to a **unique user ID**
- No sensitive data is directly stored in QR
- Data is fetched securely from backend

Summary

The Profile page ensures:

- Easy user management
- Fast hospital check-in
- Reduced manual work at hospital